Kwame Nkrumah University of
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ME 281 – Engineering Materials

Lecture 8: Dislocations and Strengthening Mechanisms

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Lecture objectives

- **Dislocations and Plastic Deformation**
  - ✓ Motion of dislocations in response to stress
  - ✓ Slip Systems
  - ✓ Plastic deformation in
    - single crystals
    - polycrystalline materials
- **Strengthening mechanisms**
  - ✓ Grain Size Reduction
  - ✓ Solid Solution Strengthening
  - ✓ Strain Hardening
- **Recovery, Recrystallization, and Grain Growth**

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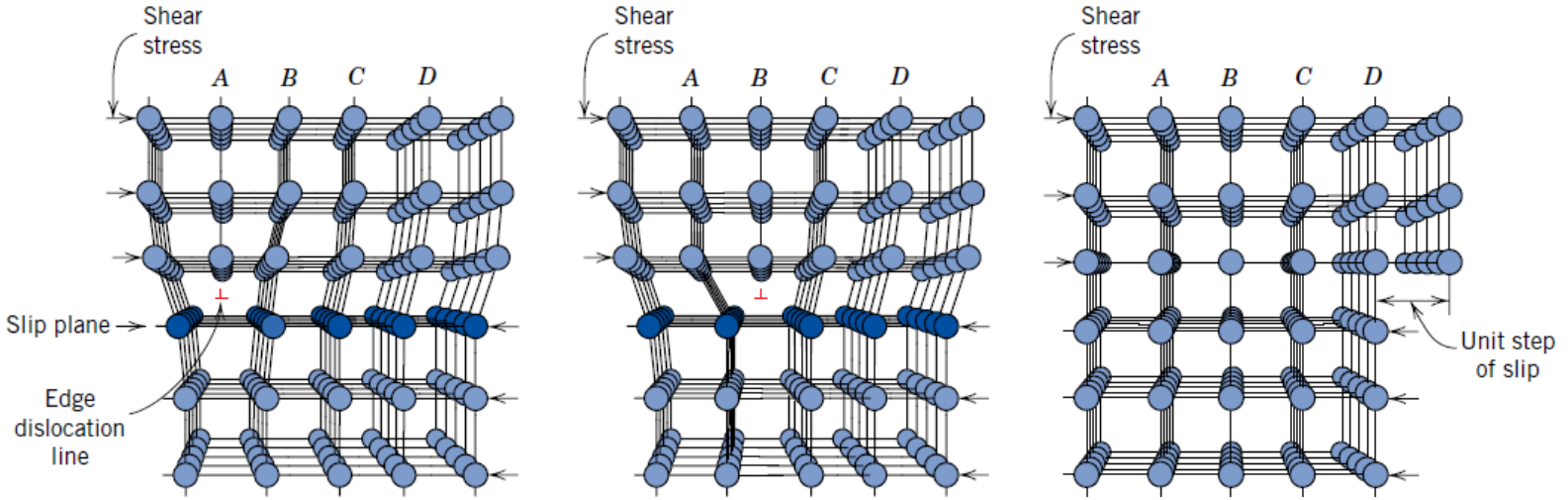
Introduction

- Two kinds of deformation exist in materials: **elastic** and **plastic**.
- Plastic deformation is permanent. A material resists plastic deformation by its strength and hardness.
- In crystalline solids, plastic deformation most often involves the motion of dislocations and linear crystalline defects.
- This chapter discusses the characteristics of dislocations, their involvement in plastic deformation and techniques for strengthening single-phase metals.

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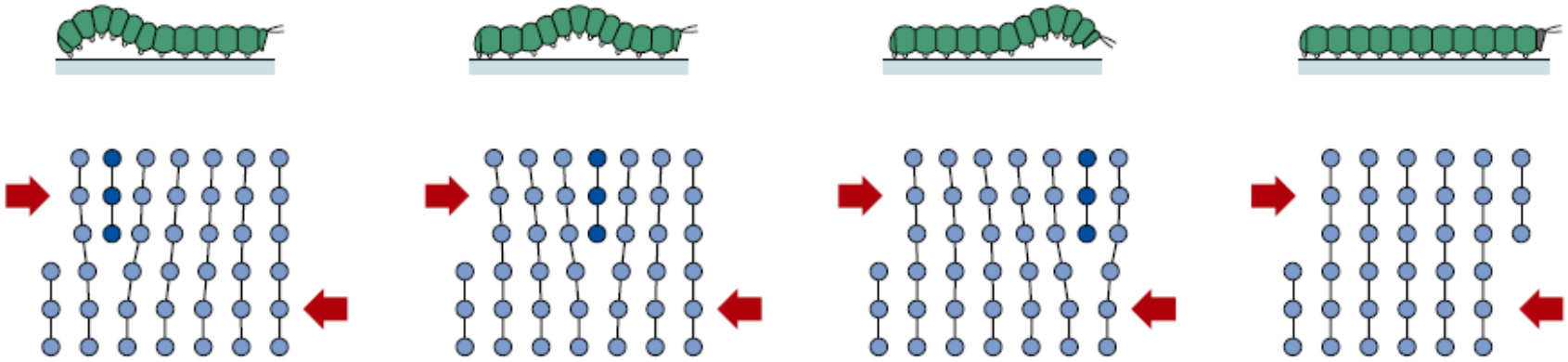
Dislocations and Plastic Deformation

- Edge and screw are the two fundamental dislocation types.
- Many dislocations in crystalline materials have both edge and screw components; these are mixed dislocations.
- **Plastic deformation corresponds to the motion of large numbers of dislocations.**

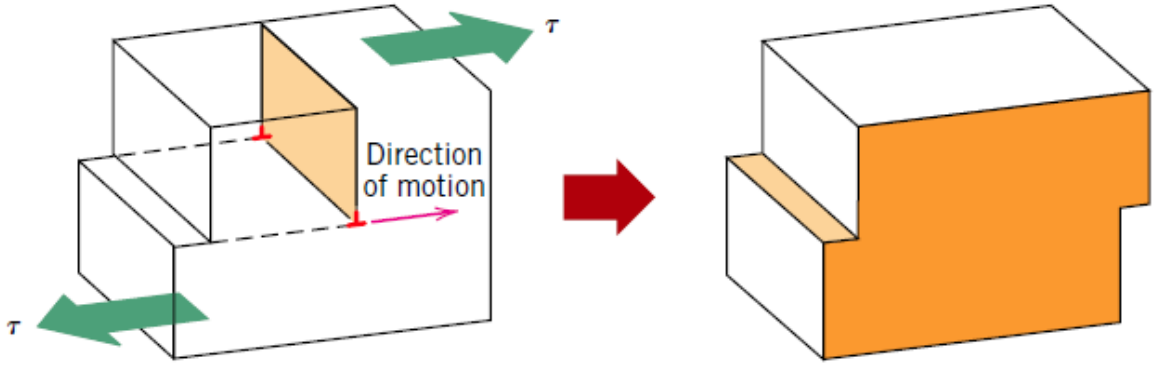


Dislocations and Plastic Deformation

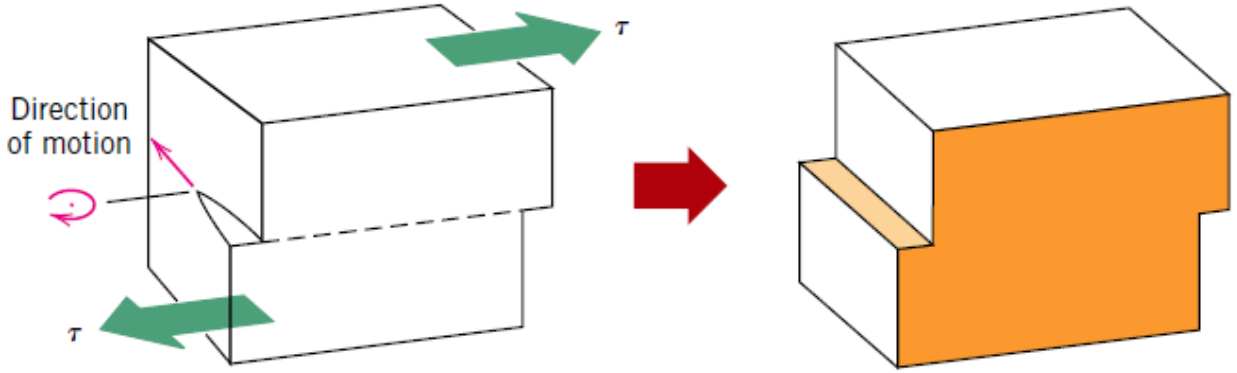
- Dislocations allow deformation at much lower stress than in a perfect crystal.
- If the top half of the crystal is slipping one plane at a time then only a small fraction of the bonds are broken at any given time and this would require a much smaller force.
- The propagation of one dislocation across the plane causes the top half of the crystal to move (**to slip**) with respect to the bottom half but we do not have to break all the bonds across the middle plane simultaneously (which would require a very large force).
- **The slip plane** – the crystallographic plane of dislocation motion.



Direction of the dislocation motion



Edge dislocation line moves parallel to applied stress



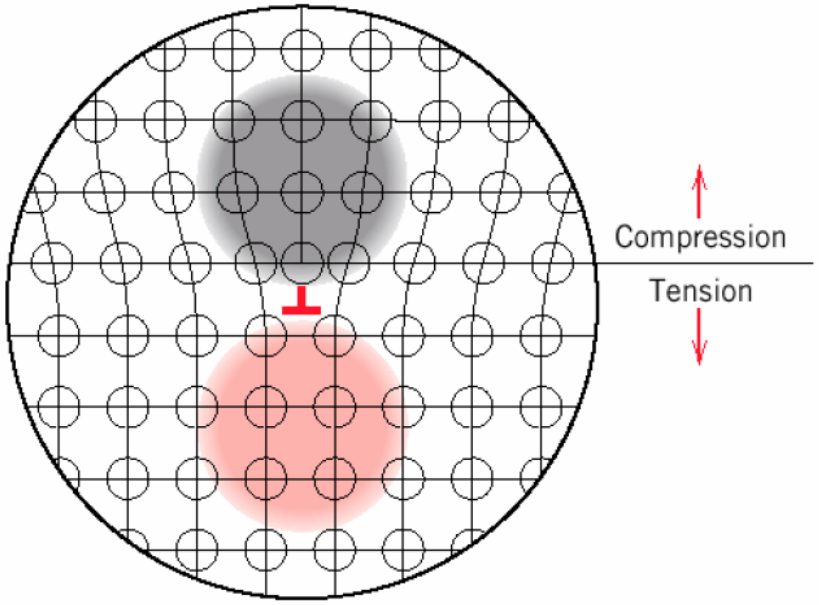
Screw dislocation line moves perpendicular to applied stress

- For mixed dislocations, direction of motion is in between parallel and perpendicular to the applied shear stress

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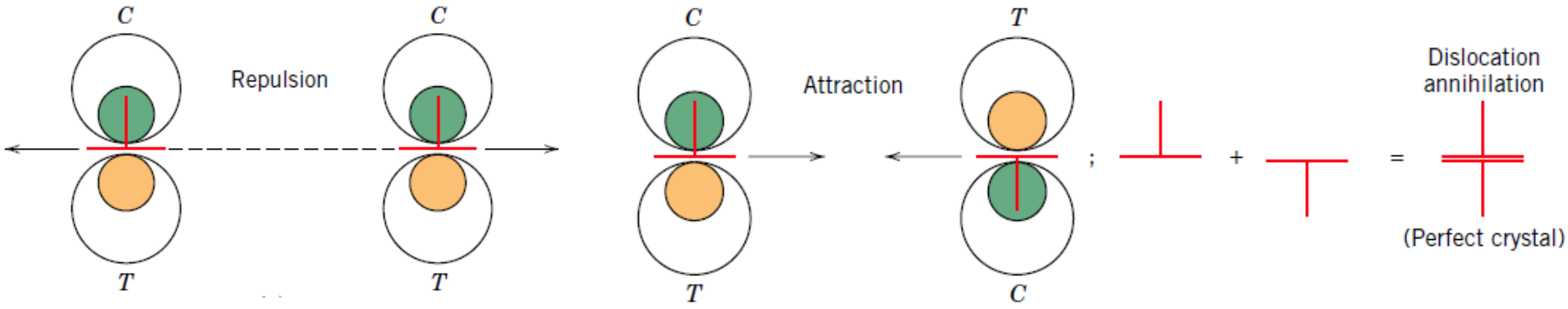
Strain field around dislocations

- Dislocations have strain fields arising from distortions at their cores - strain drops radially with distance from the dislocation core.
- **Edge dislocations introduce compressive, tensile, and shear lattice strains, screw dislocations introduce shear strain only.**



Interactions between dislocations

- The strain fields around dislocations cause them to **interact** (exert force on each other).
- When they are in the same plane, they repel if they have the same sign (direction of the Burgers vector) and attract/annihilate if they have opposite signs.



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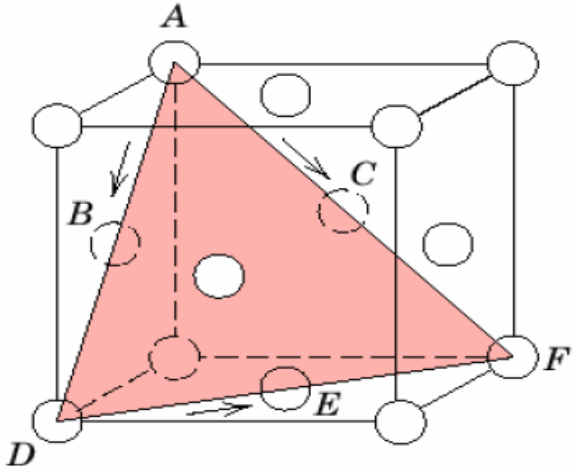
Where do dislocations come from ?

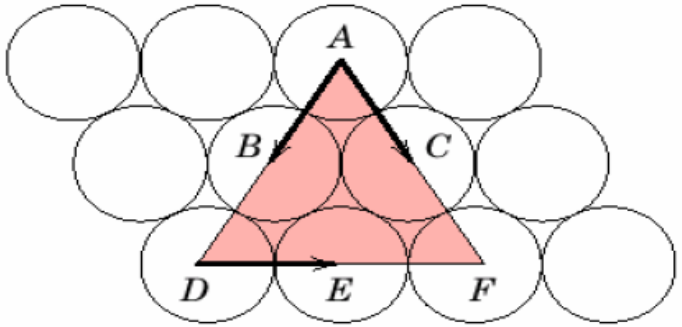
- The number of dislocations in a material is expressed as the **dislocation density** - the total dislocation length per unit volume or the number of dislocations intersecting a unit area.
- Dislocation densities can vary from $10^5\ cm^{-2}$ in carefully grown metal crystals to $10^{12}\ cm^{-2}$ in heavily deformed metals.
- **Most crystalline materials, especially metals, have dislocations in their as-formed state, mainly as a result of stresses (mechanical, thermal...) associated with the forming process.**
- The number of dislocations increases dramatically during plastic deformation. **Dislocations spawn from existing dislocations, grain boundaries and surfaces.**

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Slip Systems

- In single crystals there are preferred planes where dislocations move (**slip planes**). Within the slip planes there are preferred crystallographic directions for dislocation movement (**slip directions**). The set of slip planes and directions constitute **slip systems**.
- **The slip planes and directions are those of highest packing density.**
- BCC and FCC crystals have more slip systems as compared to HCP, there are more ways for dislocation to propagate thus, FCC and BCC crystals are more ductile than HCP crystals.

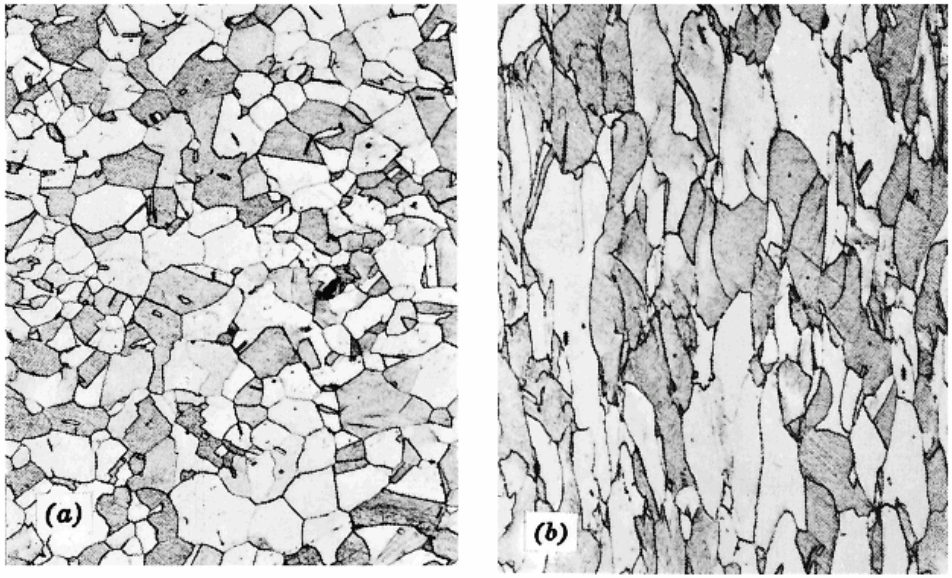




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Plastic Deformation of polycrystalline materials

- Grain orientations with respect to applied stress are random.
- The dislocation motion occurs along the slip systems with favourable orientation (i.e. that with highest resolved shear stress).
- Larger plastic deformation corresponds to elongation of grains along direction of applied stress.



Before **After**

Plastic Deformation of polycrystalline materials

- Slip directions vary from crystal to crystal
- Dislocations cannot easily cross grain boundaries because of changes in direction of slip plane and atomic disorder at grain boundaries
- **As a result, polycrystalline metals are stronger than single crystals** (the exception is the perfect single crystal without any defects, as in whiskers)

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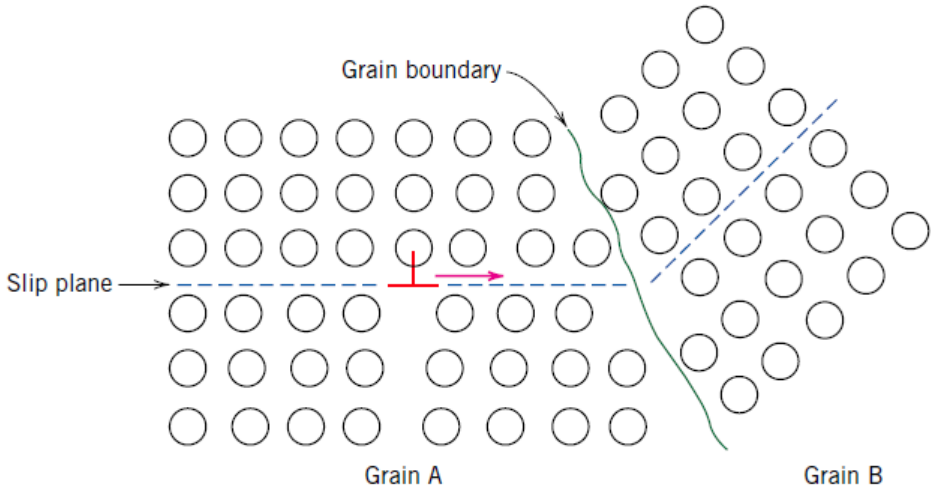
Strengthening

- **The ability of a metal to deform depends on the ability of dislocations to move**
- **Restricting dislocation motion makes the material stronger.**
- Mechanisms of strengthening in single-phase metals:
  - grain-size reduction
  - solid-solution alloying
  - strain hardening
- **Ordinarily, strengthening reduces ductility**

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Strengthening by grain-size reduction

- Grain boundary barrier to dislocation motion: slip plane discontinues or change orientation.
- Small angle grain boundaries are not very effective in blocking dislocations.
- High-angle grain boundaries block slip and increase strength of the material. A stress concentration at end of a slip plane may trigger new dislocations in an adjacent grain.



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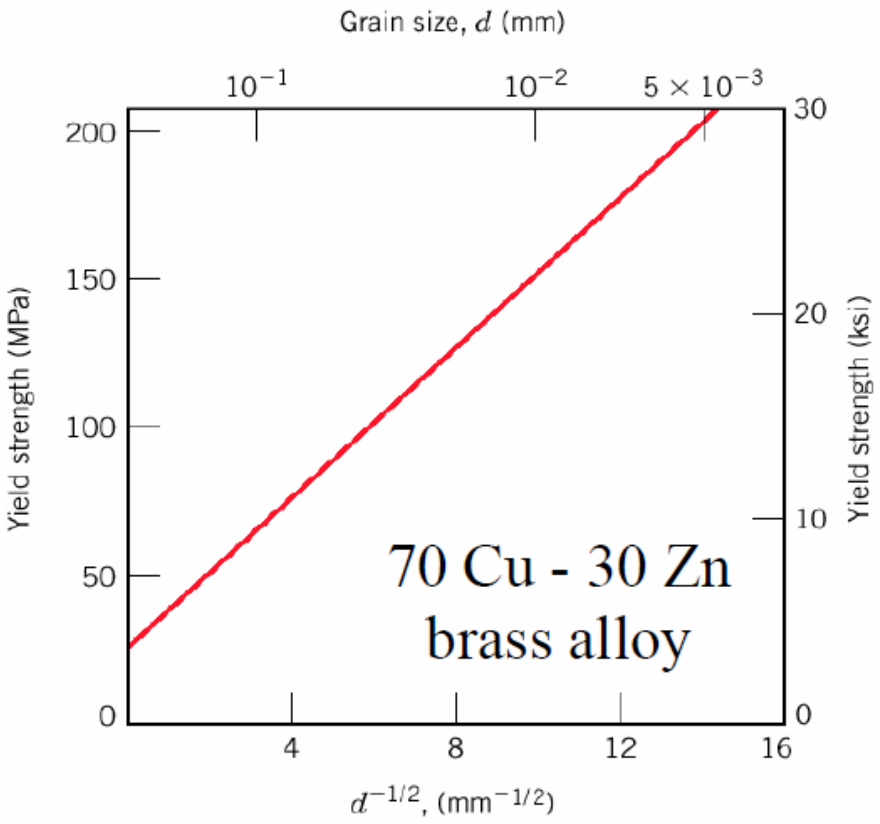
Strengthening by grain-size reduction

- The finer the grains, the larger the area of grain boundaries that impedes dislocation motion.
- Grain-size reduction usually improves toughness.
- Usually, the yield strength varies with grain size ***d*** according to Hall-Petch equation:

$$\sigma_y = \sigma_0 + k_y / \sqrt{d}$$

where $\boldsymbol{\sigma_0}$ and $\boldsymbol{k_y}$ are constants for a particular material, **d** is the average grain diameter.

- Grain size **d** can be controlled by the rate of solidification, by plastic deformation and by appropriate heat treatment.



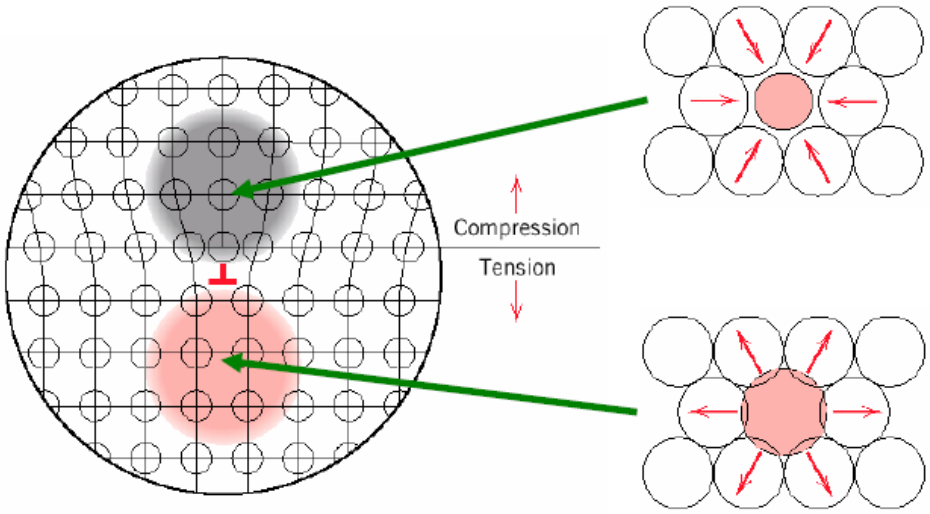
Solid-Solution Strengthening

- **Alloys are usually stronger than pure metals**
- Interstitial or substitutional impurities cause lattice strain. As a result, these impurities interact with dislocation strain fields and **hinder dislocation motion**.
- Impurities tend to diffuse and **segregate around dislocation cores** to find atomic sites that suit their radii. This reduces the overall strain energy and "anchors" the dislocations.
- Motion of a dislocation core away from the impurities moves it to a region of lattice where the atomic strains are greater (i.e. the dislocation strains are no longer compensated by the impurity atoms).

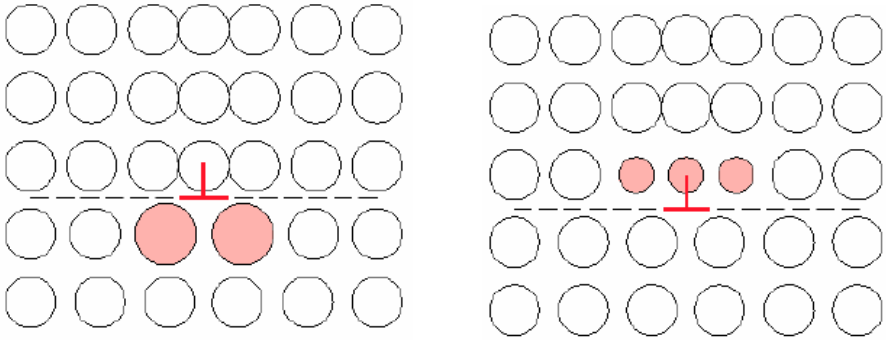
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Solid-Solution Strengthening

- Smaller and larger substitutional impurities tend to diffuse into strained regions around dislocations, leading to partial cancellation of impurity-dislocation lattice strains.



Tensile strains



Compressive strains

Strain Hardening

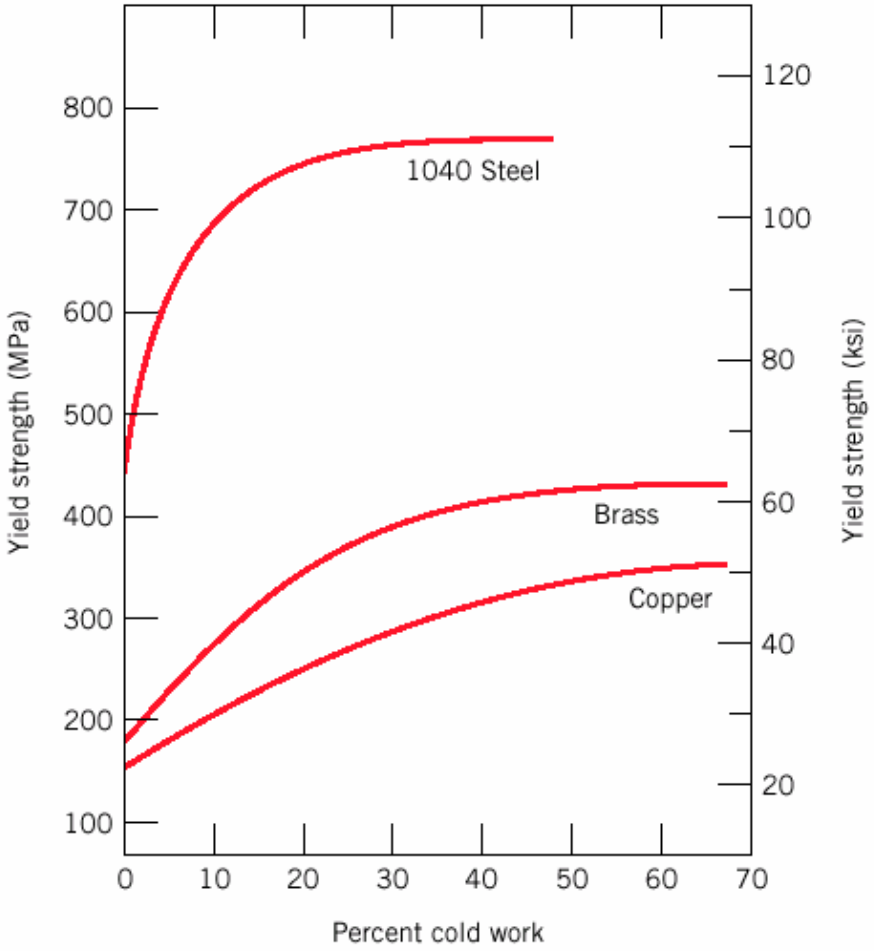
- Strain hardening is when ductile metal become stronger when they are deformed plastically.
- Strain hardening is also called **Work Hardening or Cold Working.**
- **The reason for strain hardening is the increase of dislocation density with plastic deformation.**
- The average distance between dislocations decreases and dislocations start blocking the motion of each other.
- The percent cold work (%CW) is often used to express the degree of plastic deformation:

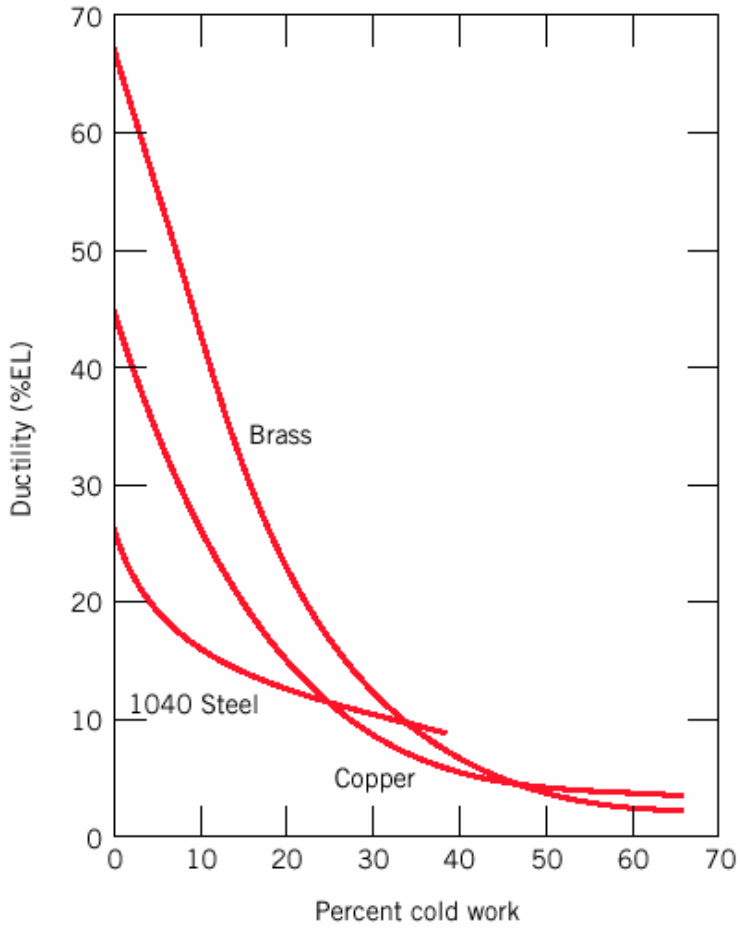
$$\%CW = \left(\frac{A_0 - A_d}{A_0} \right) \times 100$$

where A_o is the original cross-section area, A_d is the area after deformation.

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Strain hardening





Yield strength and hardness are increasing as a result of strain hardening but **ductility is decreasing** (material becomes more brittle).

Recovery, Recrystallization, and Grain growth

- **Plastic deformation increases dislocation density (single and polycrystalline materials) and changes grain size distributions (polycrystalline materials).**
- This corresponds to stored strain energy in the system (dislocation strain fields and grain distortions).
- When applied external stress is removed - most of the dislocations, grain distortions and associated strain energy are retained.
- Restoration to the state before cold-work can be done by heat-treatment and involves two processes: **recovery** and **recrystallization**. These may be followed by **grain growth**.

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Recovery

- Heating → increased diffusion → enhanced dislocation motion → decrease in dislocation density by annihilation, formation of low-energy dislocation configurations → relieve of the internal strain energy

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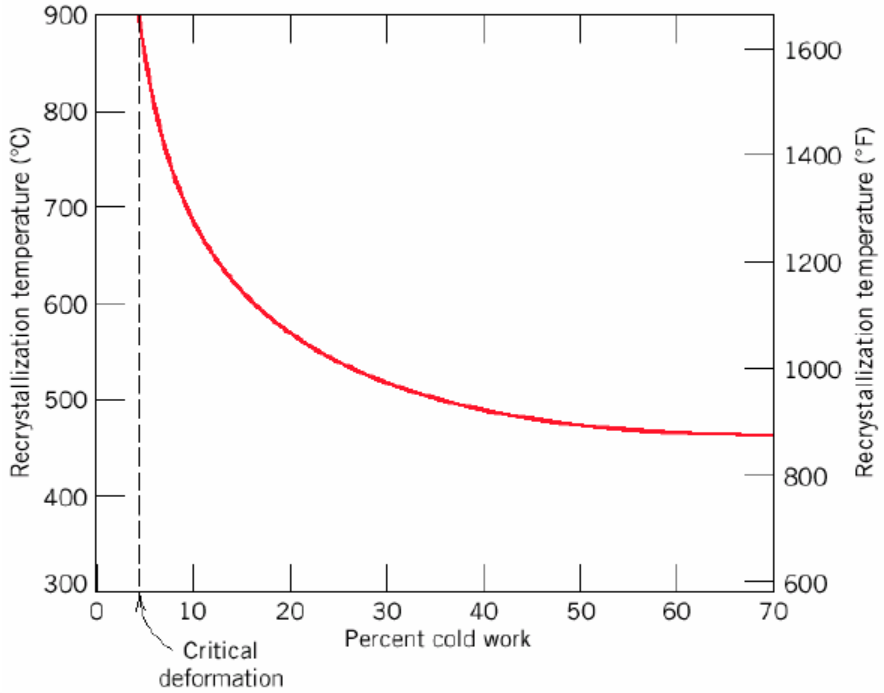
Recrystallization

- Even after recovery the grains can be strained. These strained grains of cold-worked metal can be replaced, upon heating, by strain-free grains with low density of dislocations.
- This occurs through **recrystallization – nucleation and growth of new grains**.
- The ***driving force*** for recrystallization is the difference in internal energy between strained and unstrained material.
- Grain growth involves short-range diffusion → the extent of recrystallization depends on both temperature and time.
- Recrystallization is slower in alloys as compared to pure metals.

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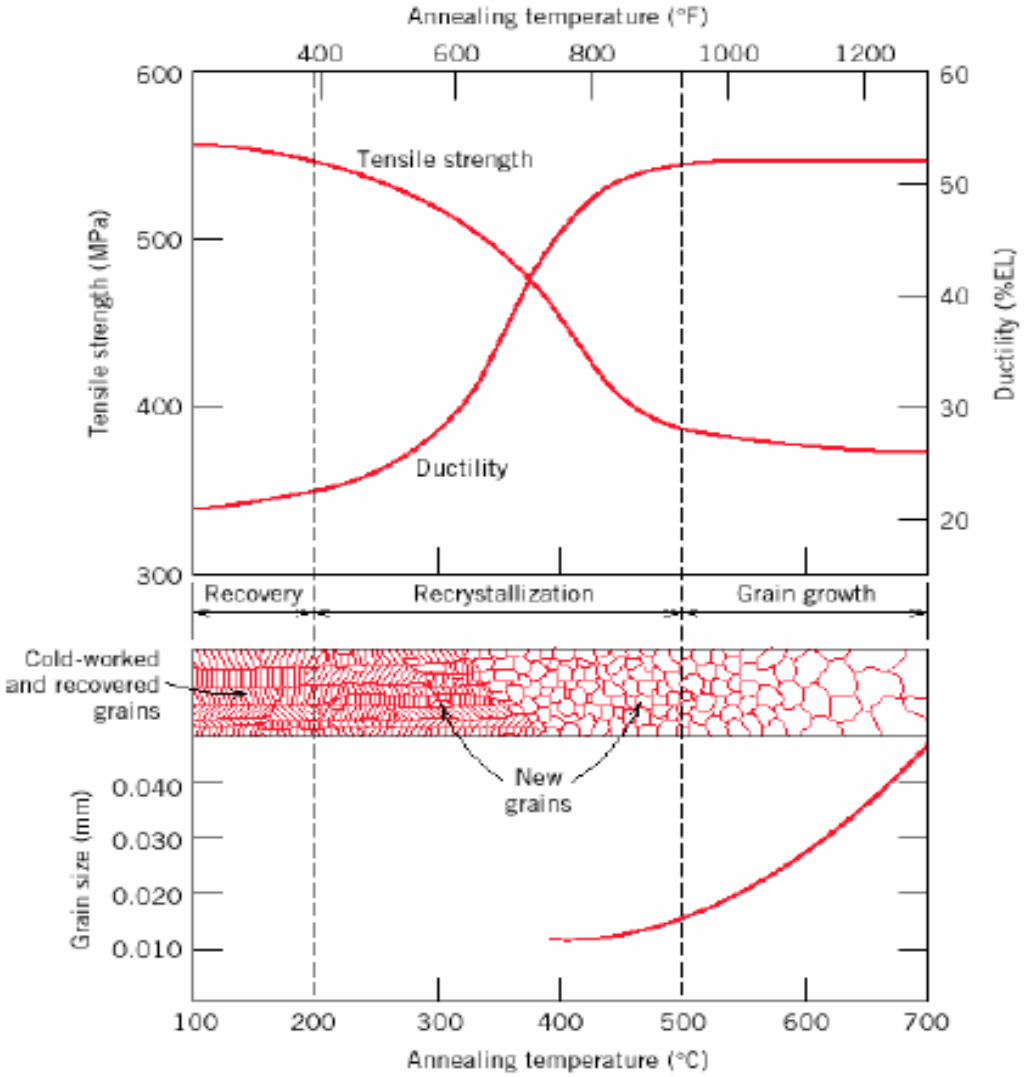
Recrystallization

- **Recrystallization temperature:** The temperature at which the process is complete in one hour. It is typically 1/3 to 1/2 of the melting temperature.
- **Recrystallization temperature increases as the %CW is decreased.** Below a "critical deformation", recrystallization does not occur.



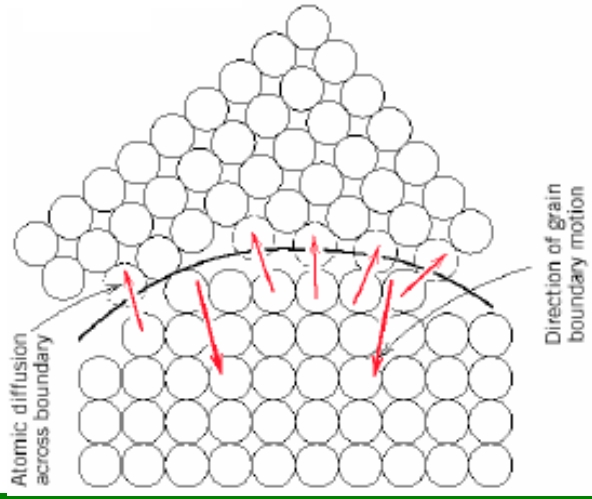
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Recrystallization



Grain growth

- If deformed polycrystalline material is maintained at annealing temperature following complete recrystallization, then further **grain growth** occurs.
- ***Driving force*** is reduction of the total grain boundary area and, hence, the energy of the system. Big grains grow at the expense of the small ones.
- Grain growth during annealing occurs in all polycrystalline materials (i.e. they do not have to be deformed or undergo recrystallization first).
- Boundary motion occurs by short range diffusion of atoms across the grain boundary → strong temperature dependence of the grain growth.



Thank You